

Week 3 : Seller and Inventory Management System

Scenario:

The E-commerce company sells product supplied by multiple sellers. The system should maintain seller details, track products supplied by each seller, and monitor inventory levels to ensure product availability.

1.Create seller and Inventory Tables

Seller Table

```
Connected.
SQL> CREATE TABLE Seller(Seller_ID VARCHAR2(10) PRIMARY KEY,Seller_Name VARCHAR2(20),Email VARCHAR2(50),Phone int,Address VARCHAR2(90));
Table created.
```

Inventory Table

```
SQL> CREATE TABLE INVENTORY(INVENTORY_ID VARCHAR(10),PRODUCT_ID VARCHAR(10),SELLER_ID VARCHAR(10),STOCK_QUANTITY INT,STOCK_STATUS VARCHAR(20),LAST_UPDATED V
ARCHAR(10));
Table created.
```

Two tables for seller and Inventory Table have been crated.

Now we have to find the relationship between the tables and apply constraints to the tables.

2.Define Relationships and Constraints

Relationships:

The **Seller** table stores information about the sellers who provide ceramic products in the e-commerce system.

- ❖ **Seller_ID** is the **Primary Key** that uniquely identifies each seller in seller table.
- ❖ It stores details such as **Seller Name, Email, Phone, and Address**.
- ❖ One seller can provide **many products**
 - **One to many relationship.**
- ❖ It helps identify which seller is responsible for selling the products.

The **Inventory** table stores information about the **quantity and availability of products**.

- ❖ Inventory_ID is the **Primary Key** in Inventory table.
- ❖ Product_ID, Seller_ID is **Foreign Key** in this table which connect other tables.
- ❖ It stores details such as **available quantity, stock level, and inventory status**.
- ❖ It helps track how many units of each ceramic product are available.

Constraints

- ✓ **PRIMARY KEY** – Uniquely identifies each record.
- ✓ **FOREIGN KEY** – Connects Product with Category.
- ✓ **NOT NULL** – Ensures a value must be entered.
- ✓ **CHECK** – Allows only valid values, such as Y or N.
- ✓ **UNIQUE** – Prevents duplicate values in a column.

1.we add primary key to Inventory Table

```
SQL> ALTER TABLE INVENTORY ADD
  2  CONSTRAINT PK_INVENTORY PRIMARY KEY(INVENTORY_ID);

Table altered.
```

We already added primary key for seller table while creating the table.

2.We added Foreign Key for Both tables:

```
SQL> ALTER TABLE INVENTORY ADD
  2  CONSTRAINT FK_PROD FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);

Table altered.

SQL> ALTER TABLE INVENTORY ADD
  2  CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID) REFERENCES Seller(Seller_ID);

Table altered.
```

We added foreign key so both seller and inventory table are connected as it has same attributes

3.Constraints to make Unique Email ID in seller table:

```
SQL> ALTER TABLE Seller ADD
  2  CONSTRAINT Un_mail UNIQUE(Email);

Table altered.
```

4.we added constraint to check the Stock quantity is Positive

```
SQL> ALTER TABLE INVENTORY ADD
  2  CONSTRAINT CK_Quen Check(Stock_Quantity >=0);

Table altered.
```

3.Insert values Seller table

```
SQL> INSERT INTO SELLER
  2 VALUES('S001','Ceramic World','ceramicworld@gmail.com',9876543210,'Chennai, Tamil Nadu');

1 row created.

SQL> INSERT INTO SELLER
  2 VALUES('S002','Clay Creations','claycreations@gmail.com',9876543211,'Madurai, Tamil Nadu');

1 row created.

SQL> INSERT INTO SELLER
  2 VALUES('S003','Modern Clay Works','modernclay@gmail.com',9876543216,'Mumbai, Maharashtra');

1 row created.

SQL> INSERT INTO SELLER
  2 VALUES('S004','Garden Ceramics','gardenceramics@gmail.com',9876543217,'Pune, Maharashtra');

1 row created.

SQL> INSERT INTO SELLER
  2 VALUES('S005','Home Clay Studio','homeclaystudio@gmail.com',9876543218,'Delhi, India');

1 row created.

SQL> INSERT INTO SELLER
  2 VALUES('S006','Traditional Pottery','traditionalpottery@gmail.com',9876543219,'Thanjavur, Tamil Nadu');

1 row created.
```

Result

```
SQL> SELECT * FROM Seller;

SELLER_ID  SELLER_NAME
-----
EMAIL
-----
PHONE
-----
ADDRESS
-----
S001      Ceramic World
ceramicworld@gmail.com      9876543210
Chennai, Tamil Nadu

S002      Clay Creations
claycreations@gmail.com      9876543211
Madurai, Tamil Nadu

SELLER_ID  SELLER_NAME
-----
EMAIL
-----
PHONE
-----
ADDRESS
-----
S003      Modern Clay Works
modernclay@gmail.com      9876543216
Mumbai, Maharashtra

S004      Garden Ceramics
gardenceramics@gmail.com      9876543217

SELLER_ID  SELLER_NAME
-----
EMAIL
-----
PHONE
-----
ADDRESS
-----
Pune, Maharashtra

S005      Home Clay Studio
homeclaystudio@gmail.com      9876543218
```

We have inserted values of seller table it contain seller detains .

4.Inventory table

```
SQL> INSERT INTO Inventory
  2 (Inventory_ID, Product_ID, Seller_ID, Stock_Quantity, Stock_Status, Last_Updated)
  3 VALUES
  4 ('INV001', 'P00001', 'S001', 120, 'In Stock', '2026-09-20');

1 row created.

SQL> INSERT INTO Inventory
  2 (Inventory_ID, Product_ID, Seller_ID, Stock_Quantity, Stock_Status, Last_Updated)
  3 VALUES
  4 ('INV002', 'P00002', 'S002', 80, 'In Stock', '2026-09-20');

1 row created.

SQL> INSERT INTO Inventory
  2 (Inventory_ID, Product_ID, Seller_ID, Stock_Quantity, Stock_Status, Last_Updated)
  3 VALUES
  4 ('INV003', 'P00003', 'S003', 70, 'In Stock', '2026-09-20');

1 row created.

SQL> INSERT INTO Inventory
  2 (Inventory_ID, Product_ID, Seller_ID, Stock_Quantity, Stock_Status, Last_Updated)
  3 VALUES
  4 ('INV004', 'P00004', 'S004', 20, 'Low Stock', '2026-09-20');

1 row created.
```

Result

```
SQL> SELECT * FROM Inventory;

INVENTORY_ PRODUCT_ID SELLER_ID STOCK_QUANTITY STOCK_STATUS LAST_UPDAT
-----
INV001      P00001      S001           120 In Stock      2026-09-20
INV002      P00002      S002           80 In Stock      2026-09-20
INV003      P00003      S003           70 In Stock      2026-09-20
INV004      P00004      S004           20 Low Stock     2026-09-20
INV005      P00005      S005           50 In Stock      2026-09-20
INV006      P00006      S006           90 In Stock      2026-09-20

6 rows selected.
```

Report 1: Seller-wise Product Report

Display: Seller Name, Number of products supplied, Available stock

```
SQL> SELECT s.Seller_Name, COUNT(i.product_ID) AS Number_of_Products,
  2 SUM(i.Stock_Quantity) AS Available_Stock
  3 FROM Seller s
  4 JOIN Inventory i ON s.Seller_ID = i.Seller_ID
  5 WHERE i.Stock_Status = 'In Stock'
  6 GROUP BY s.Seller_name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
Ceramic World	1	120
Clay Creations	1	80
Modern Clay Works	1	70
Home Clay Studio	1	50
Traditional Pottery	1	90

Report 2-Stock Availability Report

Display: Product name, Stock quantity, Stock status.

```
SQL> SELECT p.Product_Name,i.Stock_Quantity, i.Stock_Status
2 FROM Product p
3 JOIN Inventory i ON p.Product_ID=i.product_ID;
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
Dinner_Plate	120	In Stock
Ceramic_Cup	80	In Stock
Garden_Planter	70	In Stock
Flower_Vase	20	Low Stock
Wall_Hanging	50	In Stock
Dessert_Plate	90	In Stock

6 rows selected.

Report 3-Inventory Analysis

Find:

Total product available

```
SQL> SELECT COUNT(*) AS Total_Available_Product
2 FROM Inventory
3 WHERE Stock_status='In Stock';
```

TOTAL_AVAILABLE_PRODUCT
5

Product out of stock

```
SQL> SELECT COUNT(*) AS Outofstock_Product
2 FROM Inventory
3 WHERE Stock_status='Low stock';
```

OUTOFSTOCK_PRODUCT
0